

Kaitlynn T. Pineda

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RESEARCH OVERVIEW

Human-centered embodied AI researcher specializing in human-robot interaction and autonomous conversational systems. My research examines how people naturally interact with conversational AI systems in the physical world and uses behavioral insights to inform system decision-making, interaction policies, and real-world deployment. I design, build, and evaluate autonomous systems through controlled human-subject experiments and long-term deployments, with a particular focus on verbal social interactions.

EDUCATION

Johns Hopkins University, Baltimore, MD

PhD Candidate in Computer Science

Advisors: [Chien-Ming Huang](#) and [Gregory D. Hager](#)

08/2021 – Present

Expected Graduation: December 2026

Johns Hopkins University, Baltimore, MD

MSE in Computer Science

05/2024

Yale University, New Haven, CT

BS in Electrical Engineering and Computer Science, Certificate in Spanish

Advisor: [Brian Scassellati](#)

08/2017 – 05/2021

RESEARCH APPOINTMENTS

Johns Hopkins University, Baltimore, MD

Graduate Research Assistant | [Intuitive Computing Lab](#) | [LCSR](#)

08/2021 – Present

- Design and deploy LLM-driven robotic systems that integrate social dialogue into physical tasks, spanning laboratory human-subject studies and in-the-wild deployments.
- Conduct multimodal and mixed-methods analysis using mixed-effects modeling, behavioral coding, facial/gaze signal processing, and qualitative-quantitative empirical evaluation.

Yale University, New Haven, CT

Undergraduate Research Assistant | [Social Robotics Lab](#)

05/2018 – 05/2021

- Developed a Unity-based interactive game interface for studies of trust and fairness in human-robot collaboration.
- Conducted behavioral analysis of children with ASD using a long-term, in-home socially assistive robot.

PUBLICATIONS

Peer-Reviewed Conference Papers

[C-4]. **K. T. Pineda**, B. Chien, A. Mishra, T. Williams, A. Guo, Z. Xiao, & C. M. Huang. *“From Small Talk to Rapport: Exploring Robot Self-Disclosure in Collaborative Tasks.”* Proceedings of the 35th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), 2026.

🏆 **IEEE Best Student Paper Award.**

[C-3]. **K. T. Pineda**, E. Brown, & C. M. Huang. *“See You Later, Alligator”: Impacts of Robot Small Talk on Task, Rapport, and Interaction Dynamics in Human-Robot Collaboration.* Proceedings of the 2025 ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2025.

🏆 **Best Paper Honorable Mention** | Top 5.25% of submissions.

[C-2]. N. Salomons, **K. T. Pineda**, A. Adéjare, & B. Scassellati. *“We Make a Great Team!”: Adults with Low Prior Domain Knowledge Learn More from a Peer Robot than a Tutor Robot.* Proceedings of the 2022 ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2022.

[C-1]. N. Tsoi, J. Connolly, E. Adéníran, A. Hansen, **K. T. Pineda**, T. Adamson, S. Thompson, R. Ramnauth, M. Vázquez, & B. Scassellati. *Challenges Deploying Robots During a Pandemic: An Effort to Fight Social Isolation Among Children.* Proceedings of the 2021 ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2021.

🏆 **Best Paper Honorable Mention.**

Peer-Reviewed Journal Articles

[J-1]. G. Ajaykumar, **K. T. Pineda**, & C. M. Huang. *Older adults' expectations, experiences, and preferences in programming physical robot assistance.* *International Journal of Human-Computer Studies*, 180, 103127, 2023.

Preprints

[M-1]. **K. T. Pineda**, A. Mahmood, & C. M. Huang. *“You Might Like It”: How People Respond to Small Talk in Human-Robot Collaboration*. arXiv:2312.07454, 2023. [Under Review]

Workshop Papers

[W-1]. **K. T. Pineda**, A. Mishra, B. Chien, A. Guo, T. Williams, Z. Xiao, & C. M. Huang. *Shaping Small Talk: Examining the Effects of Robot Disclosure in Collaborative Settings*. InterAI Workshop: Interactive AI for Human-Centered Robotics (HRI), 2026.

INDUSTRY EXPERIENCE

Inflection AI, Palo Alto, CA

06/2026 – 08/2026

Member of Technical Staff Intern | [Inflection AI Labs](#)

- Researched context-aware AI experiences combining physical-world sensing, conversational AI, and human-centered interaction
- Designed human-subject experiments evaluating how contextual signals can inform proactive AI behavior and conversation initiation.
- Built software and hardware prototypes for experimental AI systems in collaboration with research and special projects teams

Nissan, Santa Clara, CA

06/2025 – 08/2025

Artificial Intelligence Intern | [NATC-SV](#)

- Developed LLM-driven conversational agents for human-vehicle interaction and contributed to a cross-team demo for Nissan Global CEO Ivan Espinosa.
- Built agentic workflows using RAG, function calling, speech, and vehicle data to support context-aware conversational experiences.
- Prototyped and deployed on-device conversational agents powered by local LLMs for real-world vehicle contexts.

Meta, Menlo Park, CA

06/2021 – 08/2021

Software Engineering Intern | Meta Quest (Oculus)

- Built a synthetic IMU trajectory generation pipeline in Reality Labs, producing artificial kinematic headset data from upper-body and head animation key-point sequences.
- Applied Monte Carlo sampling for spatial optimization; MATLAB to visualize and validate 3D trajectory alignment algorithms.

Software Engineering Intern | FAIAR

06/2020 – 08/2020

- Worked on dialog policy for conversational AI used with Meta Ray-Ban Glasses and built Android/web testing tools for device-to-cloud photo and video pipelines.

AWARDS & FELLOWSHIPS

[15] IEEE Best Student Paper Award, IEEE RO-MAN [C-4]	2026
[14] KROS Interdisciplinary Research Award in Social Human-Robot Interaction (nominee), RO-MAN	2026
[13] RO-MAN Student Travel Grant (\$1,000)	2026
[12] Professor Joel Dean Excellence in Teaching Award, Johns Hopkins Computer Science	2025
[11] Graduate Student Employee of the Year (nominee), Johns Hopkins University	2025
[10] Best Paper Honorable Mention, ACM/IEEE HRI [C-3]	2025
[9] IEEE Robotics and Automation Society Travel Support Award (HRI)	2025
[8] Robotics Science and Systems Inclusion Fellow	2022
[7] Computer Science Departmental Fellowship, Johns Hopkins University	2021 – 2022
[6] Howard and Jacqueline Chertkof Endowed Fellowship, Johns Hopkins University	2021 – 2022
[5] Best Paper Honorable Mention, ACM/IEEE HRI [C-1]	2021
[4] Science, Technology and Research Scholars (STARS) II Program, Yale College	2019 – 2021
[3] Alan S. Tetelman 1958 Fellowship for International Research in the Sciences, Yale College	2019
[2] Science, Technology and Research Scholars (STARS) I Summer Program, Yale College	2018
[1] Science, Technology and Research Scholars (STARS) I Academic Year Program, Yale College	2017 – 2018

TEACHING EXPERIENCE

Head Teaching Assistant, Johns Hopkins University – [EN.601.490/690 Human-Computer Interaction](#) Fall '22, '23, '24; Spring '25

Office hours, in-class activities, exam development, grading, and lecture on human-AI interaction.

Teaching Fellow / Lecturer, Yale University – CPSC 470/570 Artificial Intelligence

Spring '22

Office hours, grading, course lecture, and exam review.

Undergraduate Learning Assistant, Yale University – CPSC 223 Data Structures

Spring '20

Peer Tutor, Yale University – CPSC 223 Data Structures

Fall '20

STUDENT ADVISING & MENTORSHIP

Johns Hopkins University Master's Students

Vivek Reddy Nalla, Robotics	05/2026 – Present
Jiaming Du, Robotics	10/2025 – Present
Kush Kumar Kushwaha, Robotics	10/2025 – Present
Jie Wang, Computer Science	10/2025 – Present
Brian Chien, Robotics IEEE Best Student Paper Award Next: AI Intern at Hitachi	02/2025 – 08/2025

Johns Hopkins University Undergraduate Students

Emilie Suri, Cognitive Science	06/2026 – Present
Anvii Mishra, Computer Science IEEE Best Student Paper Award 2026 PURA Winner	11/2024 – Present
Toluwani Williams, Electrical Engineering IEEE Best Student Paper Award	11/2024 – 05/2026
Angela Guo, Computer Science IEEE Best Student Paper Award	09/2025 – 05/2026
Ethan Brown, Neuroscience Best Paper Honorable Mention Next: JHU HAMR Lab	01/2024 – 09/2024

CONFERENCE & WORKSHOP SERVICE

Peer Reviewer, ACM/IEEE International Conference on Human-Robot Interaction	HRI '24, '25, '26
Assistant Local Arrangements Chair, ACM/IEEE International Conference on Human-Robot Interaction	2027
Organizer, RSS Workshop: Close-Proximity Human-Robot Collaboration – Challenges and Opportunities	2022

UNIVERSITY SERVICE & LEADERSHIP

Johns Hopkins LCSR Graduate Student Association, President	01/2024 – 05/2026
• Lead an executive board managing \$8,000/year in student resources.	
Johns Hopkins Computer Science Graduate Student Council, Social Committee	04/2022 – 05/2026
Yale Computer Science Departmental Student Advisory Committee, Board Member	01/2020 – 05/2020

PROFESSIONAL MEMBERSHIPS & DEVELOPMENT

Cientifico Latino Graduate Student Mentorship Initiative (GSMI)	08/2020 – 05/2021
Cientifico Latino Graduate Student Engagement & Community (GSEC)	08/2021 – 05/2022
Yale Society of Women Engineers, Vice President	08/2019 – 05/2020

PRESS & MEDIA

Johns Hopkins Whiting School of Engineering News – Making Robots More Human	04/17/2025
Johns Hopkins Whiting School of Engineering News – Celebrating 2025 department awardees	05/09/2025

TECHNICAL SKILLS

Programming: Python (PyTorch, NumPy, pandas), C, C++, C#, MATLAB, JavaScript, HTML, CSS, Linux, LaTeX, Git

AI / LLM Systems: Ollama, LangChain, Hugging Face, OpenAI API (ChatGPT), Gemini API, Claude, Local LLMs, Retrieval-Augmented Generation (RAG), Vector Databases (FAISS), Function Calling, Agentic Workflows, Prompt Engineering

Speech & Multimodal Systems: Speech-to-Text (Whisper, Google Cloud), Text-to-Speech (Google Cloud), Azure Kinect depth sensing, multimodal sensing, Pupil Labs Invisible (eye tracking), OpenFace

Robotics-Hardware: Franka Emika Robot, Kinova Gen3, UR5, Fourier GR3, Arduino, Raspberry Pi, 3D printing, rapid prototyping

Robotics-Software: ROS, Gazebo, RViz, MoveIt

Tools: Unity, AutoCAD, Adobe Illustrator, Figma, Pupil Labs Invisible (eye tracking)

Data & Evaluation: Human-Robot Interaction (HRI), Human-Computer Interaction (HCI), Human-Centered Design, Human-Subject Experimentation, mixed-methods evaluation, quantitative analysis, qualitative analysis, Statistical Analysis (JMP, R), coding (MAXQDA)